The galaxy-bias model of the lssty1xdesi paper

One model and one prior for DESI C_ℓ^{gg} in thin redshift bins

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1 Summary

The question. What does DESI galaxy clustering, measured as angular power spectra C_ℓ^{gg} in thin redshift bins, constrain on its own, with a bias model a referee accepts? Thin bins resolve the growth of structure along the line of sight only if the bias of neighboring bins is not free bin by bin; tying the bins too tightly instead turns assumptions about the bias into cosmology.

The model. The galaxy power spectrum is one-loop, with seven bias quantities per bin: the linear, quadratic and tidal biases, the counterterms α_0 and α_2, and the stochastic terms SN_0 and SN_2. Each of them is its group's value, which follows calibrations where they exist, plus, depending on the quantity, a free tilt in redshift across the group and a residual of the bin with a Gaussian prior correlated in redshift (Section 3). One principle sets the priors: slicing a wide DESI sample into thin bins adds measurements but no knowledge of the bias, so the group value and the tilt carry exactly the freedom of the wide sample, and the residual is what the slicing may add within it (Section 4).

The result. On the two thin layouts, $\mathrm{FoM}(w_0, w_a)$ of DESI C_ℓ^{gg} alone grows from 0.411 and 0.244 at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ to 7.54 and 8.43 at $0.2\,h\,\mathrm{Mpc}^{-1}$, below the 17.3 of LSST Y1 3x2pt alone; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ stays below LSST Y1 as well (Section 5). Truths inside the model's priors, each holding one bias term at a full prior width in every bin, move the fit by at most 1.09σ; the largest shifts come from the line-of-sight counterterm α_2, whose prior the constraint leans on at small $k_{\max}$. Over random bias histories, the known drifts plus scatter drawn from the prior, the quoted error bars are honest at every cut and on every layout.

What the grouping is worth. A per-bin prior that looks looser than the model gains information by accident, from the slicing alone, up to a factor of 2.2 in $\mathrm{FoM}(w_0, w_a)$. Untied bins that keep the freedom of the wide samples cost a factor of 1.17 to 1.25 in $\mathrm{FoM}(w_0, w_a)$ on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ with independent per-bin terms, and up to 3.2 with residuals correlated in redshift (Section 6). One constant value of every bias quantity per wide sample claims up to 8.6 times the model's $\mathrm{FoM}(w_0, w_a)$, with error bars up to 6.2 times too small.

2 Data and scale cut

The data are the DESI angular galaxy power spectra C_ℓ^{gg} in 53 bands of width $\Delta\ell = 10$ over $15 \le \ell \le 544$, with ℓ the multipole: the auto spectrum of every redshift bin and the cross spectra of neighboring and overlapping bins. On $\Delta d_{\mathrm{com}} = 62$ Mpc these are 107 pairs: the 97 of the DESI forecast's pair list and ten LRG3×ELG1 pairs at the same redshift, whose Fingers-of-God-blurred kernels overlap (`notes/cross_correlations`). The galaxies form six tracer groups g, each a wide sample selected as a whole: BGS (the Bright Galaxy Survey), LRG1–3 (luminous red galaxies) and ELG1–2 (emission-line galaxies). Four layouts slice them: $\Delta d_{\mathrm{com}} = 62$ Mpc, 76 bins of equal

comoving width, and $\Delta z = 0.02$, 0.05 and 0.2, with 90, 36 and 10 bins of fixed redshift width. The first two are the thin layouts: a bin of comoving width $\Delta\chi$ resolves line-of-sight wavenumbers up to $k_\parallel \simeq \pi/\Delta\chi$, about $0.075\,h\,\mathrm{Mpc}^{-1}$ for the 62 Mpc bins, so thin bins keep much of the three-dimensional information [1]. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$.

Non-Limber spectra. The Limber approximation evaluates the power spectrum only at the transverse wavenumber $(\ell + 1/2)/\chi$, with χ the comoving distance, and misses the line-of-sight modes that a thin bin with sharp edges passes. The exact projection keeps them: between bins i and j, with $W_i(\chi) = n_i(z)\,dz/d\chi$ the radial kernel of bin i, n_i its normalized redshift distribution, j_ℓ the spherical Bessel function and $P(k;\,z_1, z_2)$ the unequal-time power spectrum of the projected field at $z_1 = z(\chi_1)$ and $z_2 = z(\chi_2)$,

$$C_\ell^{ij} \;=\; \frac{2}{\pi}\int d\chi_1\, W_i(\chi_1) \int d\chi_2\, W_j(\chi_2) \int_0^\infty dk\, k^2\, P(k;\,z_1, z_2)\, j_\ell(k\chi_1)\, j_\ell(k\chi_2). \tag{1}$$

The double Bessel integral is the cost: $j_\ell(k\chi_1)\, j_\ell(k\chi_2)$ oscillates and beats, and quadrature over it is slow and unstable.

Fang et al. [2] (FKEM) remove that cost by splitting the spectrum into its linear part and a nonlinear remainder. The linear part is separable, $P_{\rm lin}(k;\,z_1, z_2) = D(z_1)\,D(z_2)\,P_{\rm lin}(k)$, with $P_{\rm lin}(k)$ the linear matter power spectrum today, so in Eq. (1) each χ integral closes around its own Bessel function and becomes a one-dimensional transform that the FFTLog algorithm computes quickly. The remainder $P_{\rm mm} - P_{\rm lin}$, with $P_{\rm mm}$ the nonlinear matter power spectrum, matters only at high k, which projects to multipoles where the Limber approximation is accurate, so it is added in Limber:

$$C_\ell \;=\; C_\ell^{\rm exact}[P_{\rm lin}] \;+\; C_\ell^{\rm Limber}[P_{\rm mm}] \;-\; C_\ell^{\rm Limber}[P_{\rm lin}]. \tag{2}$$

FKEM projects three transfer kernels per galaxy bin: density, redshift space and magnification. The density and redshift-space (Kaiser) parts together give, with $b_{1,i}$ the bin's linear galaxy bias and $f = d\ln D/d\ln a$ the linear growth rate (a the scale factor),

$$\Delta_\ell^i(k) \;=\; \int d\chi\, W_i(\chi)\, D(z)\, \big[\, b_{1,i}\, j_\ell(k\chi) \;-\; f(z)\, j_\ell''(k\chi)\,\big]\,: \tag{3}$$

each factor $f\mu^2$ of the flat-sky Kaiser power, with μ the cosine of the angle to the line of sight, becomes a $-j_\ell''$ in the exact projection. In the Limber terms of Eq. (2) the identity $j_\ell''(x) = [\ell(\ell-1)/x^2 - 1]\, j_\ell(x) + (2/x)\, j_{\ell+1}(x)$ writes the second derivative with the orders ℓ and $\ell+1$, and Limber evaluates each order at its own distance, $\chi_\ell = (\ell + 1/2)/k$ and $\chi_{\ell+1} = (\ell + 3/2)/k$, so the Limber redshift-space kernel combines the bin's radial kernel at two neighbouring distances: redshift-space distortions move galaxies between neighbouring shells. The third kernel, lensing magnification, is proportional to the magnification bias $b_{\rm mag}$, the response of the observed number density to the lensing convergence; this forecast sets $b_{\rm mag} = 0$ in every DESI bin and every LSST Y1 lens bin, so magnification is neglected.

For thin bins the split of Eq. (2) is not enough. Limber's error shrinks with the width of the radial kernel, so for a thin bin the Limber remainder stays wrong far past the switch multipole: the hand-off fails exactly where accuracy was the goal. The density–density part of every DESI spectrum – the linear-bias term, the one-loop terms and the α_0 counterterm of Eq. (5) – is therefore projected by a low-rank factorization instead. Per bin, a singular value decomposition writes the square root of the real-space galaxy power spectrum $P_g(k, z)$ as a sum of five products of a function of k and a function of z, each projected with its own FFTLog transform; the unequal-time spectrum

is the geometric mean $\sqrt{P_g(k,z_1)}\,\sqrt{P_g(k,z_2)}$, exact at equal times and treating the two fields as fully correlated otherwise. This carries the full one-loop power with no Limber hand-off up to a multipole cap at $\ell = 1000$, below which every kept band lies.

The hybrid in one sentence: the density–density part runs through the low-rank factorization; the Kaiser terms and the α_2 counterterm keep the FKEM split, projected exactly with $P_{\rm lin}$ and corrected in Limber by Eq. (2); the stochastic terms are projected as exact templates; and magnification is neglected. The one-loop power is not pushed through the velocity terms because velocities do not grow like the density field once nonlinear.

Parameters and figures of merit. Seven cosmological parameters vary with flat priors: $10^9 A_s$, with A_s the amplitude of the primordial curvature power spectrum; the spectral index n_s; the Hubble constant H_0; the baryon and matter densities today, Ω_b and Ω_m, in units of the critical density; and w_0 and w_a of the dark-energy equation of state $w_0 + w_a(1-a)$, with a the scale factor. Seven Fingers-of-God velocity dispersions σ_v, which smooth the bins' redshift distributions, vary under Gaussian priors. With σ a parameter's marginalized one-sigma error, $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) ellipse, and $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way. The reference is LSST Y1 3x2pt alone [3].

The scale cut. Matter is less nonlinear at higher redshift, so bin i is cut at $k_{\rm max}/D(\bar{z}_i)$, and every $k_{\rm max}$ here is the $z = 0$ value. The cut is the non-Limber UV-fraction criterion of Saraivanov et al. [4], after DeRose and Chen [5]: each band power splits into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; 0.05 \tag{4}$$

and within the shipped mask, which stops every bin at $0.176\,h\,\mathrm{Mpc}^{-1}$. The cut is built on linear-bias spectra. The forecasts scan $k_{\rm max} = 0.05, 0.075, \ldots, 0.2\,h\,\mathrm{Mpc}^{-1}$.

The covariance. The band-power covariance is Gaussian, computed at the one-loop fiducial (every bin's linear bias on its group's trend, b_2 and b_{s^2} on their calibrations, every other bias term zero) for 14,000 square degrees, with its inverse rescaled to the forecast's 12,300. The data vectors are the model's own prediction at that fiducial.

3 The bias model

The one-loop power spectrum. The galaxy power spectrum of bin i, in redshift space, is

$$\begin{aligned} P_{g,i}(k,\mu) \;&= b_{1,i}^2\, P_{\rm mm}(k) + P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) + \alpha_{0,i}\, k^2 P_{\rm mm}(k) \\ &\quad + \big(2\, b_{1,i} f\mu^2 + f^2\mu^4\big) P_{\rm lin}(k) + \big(b_{1,i} + f\mu^2\big) f\, \alpha_{2,i}\, \mu^2 k^2 P_{\rm lin}(k) \\ &\quad + \frac{1}{\bar{n}_i}\Big(\mathrm{SN}_{0,i} + \mathrm{SN}_{2,i}\, f_{\rm sat}\sigma_{\rm sat}^2\, k^2\mu^2\Big), \end{aligned} \tag{5}$$

with k the comoving wavenumber, μ the cosine of the angle to the line of sight, $P_{\rm mm}$ the nonlinear (halofit) matter power spectrum, $P_{\rm lin}$ the linear one, $f = d\ln D/d\ln a$ the growth rate and $\bar{n}_i$ the bin's number density. $P_{\rm loop}$ holds the one-loop terms of the quadratic bias b_2 and the tidal bias b_{s^2}, the responses of the galaxy density to the squared density contrast and to the squared tidal field [6, 7]; the third-order bias is zero. α_0 is the k^2 counterterm, which absorbs the sensitivity

of the loops to small scales. The $P_{\rm lin}$ terms are linear (Kaiser) redshift-space distortions on each bin's Fingers-of-God-convolved redshift distribution, with no redshift-space one-loop terms, and α_2 is the line-of-sight counterterm in the convention of the DESI 2024 full-shape analysis [8], in which $\alpha_2 k^2$ is a fractional correction to the velocity part of the Kaiser power. Equation (5) writes these terms with $P_{\rm lin}$; their projection adds the Limber difference between $P_{\rm mm}$ and $P_{\rm lin}$ (Section 2), so where the Limber approximation holds they carry the nonlinear power. Both counterterms are in $h^{-2}\,\mathrm{Mpc}^2$. SN_0 is the white noise beyond Poisson, in units of the Poisson power $1/\bar{n}_i$ that the covariance carries, and $\mathrm{SN}_2\,k^2\mu^2$ the stochastic term of the satellites' velocities, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$, with $f_{\rm sat}$ the satellite fraction and $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units: $(0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG and $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG [8], and the LRG values for BGS.

One form for every bias quantity. Every bias quantity X of Eq. (5), in bin i of group g, is

$$X_i \;=\; X_g(\bar{z}_i) \;+\; t_{X,g}\,(\bar{z}_i - \bar{z}_g) \;+\; \varepsilon_{X,i}, \tag{6}$$

with $X_g(\bar{z}_i)$ the group value of X, which includes the calibration trend where X has one, $t_{X,g}$ a tilt in redshift across the group, $\bar{z}_g$ the mean redshift of the group's bins, and $\varepsilon_{X,i}$ a residual of the bin. The group value and the tilt are the freedom of the wide sample; the residual is what thin slicing may add within it (Section 4). Table 1 lists what each quantity carries.

For the linear bias the three parts enter as fractions of the group value $B_g/D(\bar{z}_i)$, which keeps the clustering amplitude $b_1 D$ constant across the group:

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{7}$$

with B_g the group amplitude, s_g the tilt, ε_i the residual and δ_i the fractional deviation from the group value. The group values of the quadratic and tidal biases are calibrations evaluated at the bin's own linear bias, plus a group offset:

$$b_{2,i} = b_2^{\rm L}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,(\bar{z}_i - \bar{z}_g), \qquad b_{s^2,i} = -\tfrac{4}{7}\,\big(b_{1,i}-1\big) + \Delta_{s,g} + t_{s,g}\,(\bar{z}_i - \bar{z}_g), \tag{8}$$

with $b_2^{\rm L}(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$ the separate-universe fit of Lazeyras et al. [9] and the second relation the co-evolution of a population formed with local bias [7]. Only the linear bias carries a residual of its own: the quadratic and tidal biases receive each bin's residual through $b_{1,i}$ in their calibrations, and the counterterms and stochastic terms have none.

Why SN_2 matters on thin bins. A constant power N projects exactly as $N\int d\chi\, W_i W_j/\chi^2$ onto bins i and j, with $W_i(\chi)$ the Fingers-of-God-convolved kernel of bin i; bins of two different samples share no stochastic term. A power proportional to $k^2\mu^2 = k_\parallel^2$ is that of the line-of-sight derivative of a white noise, so SN_2 projects with each kernel replaced by its derivative, $W_i' = dW_i/d\chi$:

$$C_\ell^{ij}\big|_{\mathrm{SN}_2} \;=\; U_{ij}\int d\chi\,\frac{W_i'(\chi)\,W_j'(\chi)}{\chi^2}, \tag{9}$$

with U_{ij} the mean of the two bins' amplitudes $\mathrm{SN}_2 f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}$. A thin bin's W_i' peaks at its two edges with opposite signs, and adjacent bins share an edge, where one kernel rises as the other falls: the projected SN_2 of neighbors is anticorrelated. Normalized by the two auto terms, the adjacent term has a median of -0.50 over the 73 adjacent pairs of one sample on $\Delta d_{\rm com} = 62$ Mpc, against $+0.05$ for SN_0. No white noise, per group or per bin, mimics this pattern, and a fit without SN_2 is biased on the thin layouts already at $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ (`notes/higher_order_bias`).

quantity	group value	tilt	residual
b_1	$B_g/D(\bar{z}_i)$; B_g flat	s_g, flat	ε_i, Eq. (10)
b_2	$b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}$; $\Delta_{2,g}$ width 2	flat	none of its own
b_{s^2}	$-\frac{4}{7}(b_{1,i}-1) + \Delta_{s,g}$; $\Delta_{s,g}$ width 1	flat	none of its own
α_0, α_2	width $0.5/k_{\rm max}^2$	flat	none
SN_0	width 2	none	none
SN_2	width 5	none	none

Table 1. What each bias quantity carries in Eq. (6). A width is that of a Gaussian prior centered on zero, so the fiducial model sits on the calibrations with no counterterm and no stochastic term beyond Poisson; flat marks a flat prior. The counterterm width is the rule of the DESI 2024 full-shape analysis [10, 8], a correction of 0.5 at the cut, evaluated at each cut ($12.5\,h^{-2}\,\mathrm{Mpc}^2$ at $0.2\,h\,\mathrm{Mpc}^{-1}$). A group of one bin has no tilt.

4 The prior: the freedom of the wide sample

The principle. DESI selects each group as one wide sample, and what is known about its bias, from calibrations to the priors of full-shape analyses, refers to that sample. Slicing it into thin bins adds measurements of the clustering, not knowledge of the bias. A thin-bin prior is therefore fair only if it leaves each group the freedom of its wide sample, whatever the slicing. In Eq. (6) that freedom is the group value and the tilt, whose priors are set for the wide sample; the residual is what the slicing may add within it, local features of the size the selection can produce, and its prior must not tighten as the bins get thinner.

The linear bias. The level B_g and the tilt s_g are free, and the residual has a Gaussian prior, zero across groups and correlated within one,

$$\langle \varepsilon_i\, \varepsilon_j \rangle \;=\; \sigma_g^2\, \exp\!\left(-\,\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{10}$$

with σ_g the width of one bin's residual and L the correlation length in redshift. The kernel defines a random process $\varepsilon(z)$ along each group before any binning, which a layout only samples at its bin redshifts: the prior on a coarse set of redshifts is the marginal of the prior on a finer one, so slicing finer adds no prior knowledge. Two neighbors at separation Δz differ by an rms $\sigma_g[2(1-e^{-\Delta z/L})]^{1/2}$. The prior charges only the residual around each group's best straight line (Section 7), so a group of one or two bins, as on $\Delta z = 0.2$, has a free bias in every bin.

No measurement resolves the bias of these samples below $\Delta z \approx 0.1$, so the sizes come from what selects them (`notes/correlated_bias_prior_v2`). Coherent changes move a whole group along a nearly straight line, which the free tilt absorbs: a conserved population, whose bias obeys $b_1 - 1 \propto 1/D$ [11], departs from B_g/D by about 1% per $\Delta z = 0.05$, and the $r < 19.5$ flux limit of BGS [12] drifts its bias by 22% across the group. Local changes, such as the 4000 Å break crossing a filter edge over $\Delta z \approx 0.03$–0.05, move bins independently. Hence $\sigma_g = 3\%$ in every group, above the largest derived departure from a line (the 2.4% rms that the BGS flux limit leaves), and $L = 0.05$, the scale of a filter-edge crossing; neighbors at $\Delta z = 0.02$ then differ by 2.4% rms. The draws of Figure 1 keep the line of the known drift and add local features of a few percent.

The higher-order terms. The calibrations of b_2 and b_{s^2} describe halo populations; a wide sample adds how far it sits from them. Each group therefore gets one offset per family, with the

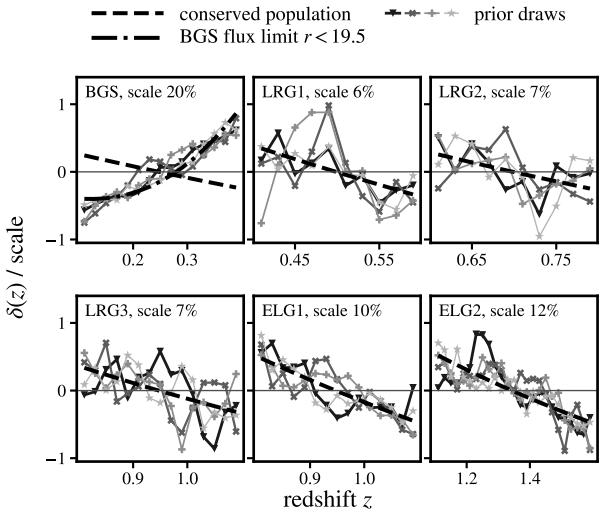


Figure 1. Four draws of the prior on the $\Delta z = 0.02$ bins, one panel per group, each panel divided by the round scale written in it. Thick black dashed: the conserved-population drift; thick black dash-dot (BGS only): the $r < 19.5$ flux-limit drift. Thin gray lines: the draws of $\delta(z)$, each with its own shade, line width and marker. Every draw keeps the best straight line through its group's reference drift, the flux limit in BGS and the conserved population elsewhere, and differs from it by a residual drawn with $\sigma_g = 3\%$ and $L = 0.05$. The level of every curve is removed, since B_g is free.

widths of Table 1: twice what the known imperfection of the calibrations requires (the scatter of the Lazeyras fit and the 30% spread between co-evolution and excursion-set tidal biases), so that the calibrations act as prior centers where the loops reach 15% of the galaxy power. A calibration of the wide sample says nothing about how the offset changes inside it, so the tilt is free; it also absorbs exactly a calibration that drifts linearly across a group. The counterterms follow the same pattern, since their scale follows the host halos of the selection. The stochastic amplitudes do not tilt: their Poisson part is known bin by bin.

5 Forecast

Figures of merit. Figure 2 draws both FoMs against $k_{\rm max}$. FoM(w_0, w_a) is 0.411, 0.244, 0.160 and 0.0496 at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ and 7.54, 8.43, 2.48 and 0.288 at $0.2\,h\,{\rm Mpc}^{-1}$, in layout order ($\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2); FoM$(\Omega_m, 10^9 A_s)$ at $0.2\,h\,{\rm Mpc}^{-1}$ is 55.9, 58.2, 21.8 and 2.97. The thin layouts carry most of the information. LSST Y1 3x2pt alone gives 17.3 and 215; DESI C_ℓ^{gg} alone stays below it in both FoMs at every cut.

A stress test. Figure 3 confronts the model with truths inside its priors, each holding one term at a full prior width with the same value in every bin: $\alpha_2 = \pm 0.5/k_{\rm max}^2$, ${\rm SN}_2 = \pm 5$, or $(\Delta_2, \Delta_s) = \pm(2, 1)$; a fourth, the calibration drift, is absorbed exactly by the tilts. A fit's shift is its linear response (Section 7), in units of its own marginalized ellipse for (w_0, w_a) and of σ for Ω_m and A_s. On $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.02$ the line-of-sight counterterm α_2 sets the largest shift at every cut from $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ on. At $0.075\,h\,{\rm Mpc}^{-1}$ it moves the fit by 0.99σ, 0.57σ and 0.24σ on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and 0.05, while the other truths move the two thin layouts by at most 0.22σ. From $0.1\,h\,{\rm Mpc}^{-1}$ on, the largest shift on the three thinner layouts lies between 0.33σ and 1.09σ; on $\Delta z = 0.2$, whose wide bins barely see α_2, the offsets set it, at most 0.22σ at any cut.

A truth inside a Gaussian prior can move the fit by about 1σ without making the error bars dishonest. With $\mathbf{p}_{\rm t}$ the truth's departure from the fiducial point and $F_{\rm prior}$ the prior precision (Section 7), the fitted departure is $\mathbf{p}_{\rm t} - F^{-1} F_{\rm prior}\,\mathbf{p}_{\rm t}$: the prior pulls the fit back toward its center. For a truth at one prior width in a single prior-held parameter α, the shift of a parameter θ, in

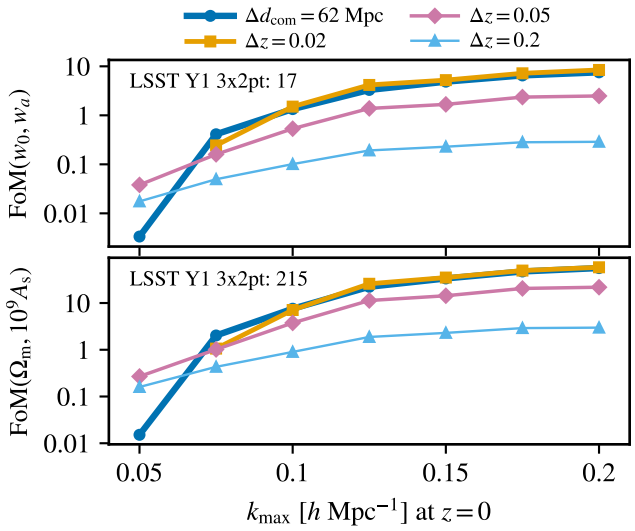


Figure 2. FoM(w_0, w_a) (top) and FoM$(\Omega_m, 10^9 A_s)$ (bottom) of DESI C_ℓ^{gg} alone against $k_{\rm max}$. Color, line width and marker: the layouts. Text in each panel: LSST Y1 3x2pt alone. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

units of its own σ, is $-\rho\,\sigma_{\alpha,\rm post}/\sigma_{\alpha,\rm prior}$, with ρ the correlation of θ and α in F^{-1} and $\sigma_{\alpha,\rm post}$ and $\sigma_{\alpha,\rm prior}$ the marginalized and prior widths of α; its size is at most 1, and the α_2 truths add such shifts over the six groups. A shift near 1σ therefore means that the constraint at that cut leans on the α_2 prior, whose width $0.5/k_{\rm max}^2$ is largest at small $k_{\rm max}$. The marginalized errors already include that prior: averaged over α_2 truths drawn from it, the shift and the data noise together have a covariance no larger than F^{-1} in any direction.

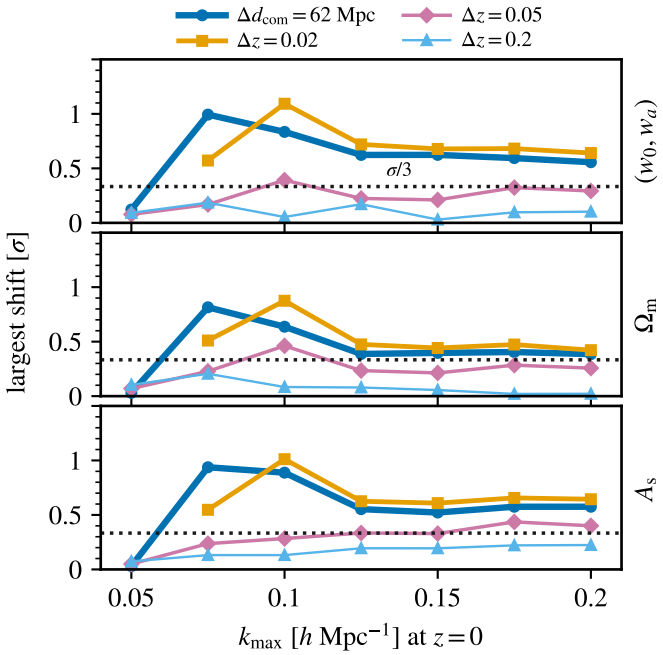


Figure 3. The largest shift of a fit over four truths, each the same in every bin: α_2, SN_2 and the offsets (Δ_2, Δ_s) at plus or minus one prior width, and the calibration drift (Δ_2 and Δ_s changing linearly by 1 and 0.5 across each group), which the tilts absorb exactly. Top: (w_0, w_a), in units of the fit's marginalized ellipse; middle and bottom: Ω_m and A_s, in units of their marginalized σ; the three panels share one linear axis from 0. Color, line width and marker: the layouts; dotted line, labeled in place: $\sigma/3$. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

The error bars are honest. How well a fit's quoted error bars cover where the fit actually lands is measured by $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, with $\chi^2 = \Delta p^{\mathsf{T}} \mathsf{C}_p^{-1} \Delta p$, Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\rm dof} = 2$ for (w_0, w_a) and 1 for Ω_m or A_s, and

$\langle\cdot\rangle$ the average over the truths and the data noise; $\eta = 1$ for honest error bars. Over random bias histories, the known drifts plus a residual drawn from the prior, $\eta \leq 1.028$ at every cut and on every layout (at most 1.018 for (w_0, w_a), 1.028 for Ω_m and 1.004 for A_s), and the 1σ region covers the truth in at least 66.9% of universes, against 68.3% for honest error bars.

6 What the grouping is worth

The principle of Section 4 says that a thin-bin prior must keep the freedom of the wide original samples, and add nothing to it. Figure 4 puts the principle to a test on six versions of the priors: five untie the bias bin by bin, and one ties it harder, to a constant across each sample. The top two rows divide each version's FoM by the model's; the bottom row asks whether its error bars are honest, with the statistic η of Section 5 for (w_0, w_a) over the same random bias histories. The group width of a parameter is the width the model gives it (Table 1), and N is the number of bins of its group. The first four versions free $b_{1,i}$ in every bin and give Δ_2, Δ_s, α_0, α_2, SN_0 and SN_2 one value per bin; the fifth keeps the model whole, its b_1 prior included; the sixth ties every bin of a sample to one value.

Why a naive per-bin prior gains. Give each of N bins its own offset, independent of the others, each with a Gaussian prior of width σ. Their average is the sum of the N offsets divided by N. Variances of independent numbers add, so the sum has variance $N\sigma^2$ and the average $N\sigma^2/N^2 = \sigma^2/N$: the average has width $\sigma/\sqrt{N}$. With $\sigma = 2$, the group width of Δ_2, and $N = 9$, the prior pins the group's average offset to 2/3, three times tighter than the wide sample knows it. Slice the same sample into more bins and the pin tightens further, with no new knowledge of the bias.

The six versions, in the words of the figure's legend:

- **per bin, group width**: looks looser than the model, but pins each group's average to the group width over $\sqrt{N}$. It lies above the model on every layout and at every cut, by up to a factor of 2.2: a gain by accident, from the slicing.
- **per bin, group width** $\times\sqrt{N}$: gives each group's average back the group width. On $\Delta z = 0.2$, whose groups hold one or two bins, it still gains, because it fixes the trend inside a group that the model leaves free.
- **per bin,** $\times\sqrt{N}$ **+ group tilt**: frees that trend too, and so restores the whole wide-sample freedom. Its prior is at least as loose as the model's in every direction (each bin's variance is N times the squared group width, against one squared group width for the shift common to all bins), so it can only lose FoM.
- **per bin, on top of the model**: keeps the model's group offsets and tilts, the wide-sample freedom, and adds an independent per-bin value of the group width to every term.
- **correlated residuals on top of the model**: keeps the model exactly and adds to each of Δ_2, Δ_s, α_0 and α_2 a per-bin residual with the prior of b_1's: Gaussian, of the term's group width, correlated in redshift over $L = 0.05$, and charged only on the deviations from the group's line. In the form of Eq. (6) it fills the residual column of Table 1 for these four terms. In a group of one or two bins the offset and the tilt already fix every bin, so on $\Delta z = 0.2$ this version is the model.

- **one constant value per wide sample**: every thin bin of a group shares one value of every bias quantity, constant in redshift. The linear bias is a plain constant, $b_{1,i} = B_g$ instead of $B_g/D(\bar{z}_i)$, and Δ_2, Δ_s, α_0, α_2, SN_0 and SN_2 take one value per group, under the model's priors on the group values, with no tilts and no residuals. It is the simplest way to slice a wide sample.

What untying the bins costs. The last two versions keep the wide-sample freedom and add only the freedom of untied bins, so they measure what untying really costs, and they differ sharply. Independent per-bin terms on top of the model cost a factor of 1.17 to 1.25 in $\mathrm{FoM}(w_0, w_a)$ on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, and 1.05 to 1.09 on the three thinner layouts at $0.2\,h\,\mathrm{Mpc}^{-1}$. Correlated residuals cost a factor of 3.2 on $\Delta z = 0.02$ at $0.075\,h\,\mathrm{Mpc}^{-1}$ and 1.36 to 1.55 on the three thinner layouts at $0.2\,h\,\mathrm{Mpc}^{-1}$; on $\Delta d_{\mathrm{com}} = 62$ Mpc their fit is not well defined at $0.075\,h\,\mathrm{Mpc}^{-1}$, and they cost 3.0 at $0.1\,h\,\mathrm{Mpc}^{-1}$. A residual that is smooth in redshift can mimic a change of the cosmology across neighboring bins, while independent per-bin terms average down over the bins of a group. On $\Delta z = 0.2$ neither version costs more than a factor of 1.1 at any cut.

What a constant per sample claims. The sixth version claims the largest gains of the figure: 3.3 to 8.6 times the model's $\mathrm{FoM}(w_0, w_a)$ over the layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, 1.9 to 3.6 at $0.2\,h\,\mathrm{Mpc}^{-1}$, and 1.4 to 5.2 times $\mathrm{FoM}(\Omega_m, 10^9 A_s)$. The bottom row prices the claim. For this version the truth also holds the evolution of the bias inside each sample, which a constant cannot follow: the fiducial bias itself evolves, the linear bias as $B_g/D(\bar{z}_i)$, and b_2 and b_{s^2} with it through the calibrations. That evolution alone shifts (w_0, w_a) by 3.8 to 6.3σ at $0.075\,h\,\mathrm{Mpc}^{-1}$ and 4.7 to 16σ at $0.2\,h\,\mathrm{Mpc}^{-1}$; the known drifts partly cancel it, which is why η comes out smaller than those shifts. The error bars are still 2.2 to 3.8 times too small at $0.075\,h\,\mathrm{Mpc}^{-1}$ and 3.2 to 6.2 at $0.2\,h\,\mathrm{Mpc}^{-1}$, where they reach 3.4 for Ω_m and 3.9 for A_s, and the 68.3% region covers the truth in 4% to 8% of universes (3.8% on $\Delta d_{\mathrm{com}} = 62$ Mpc). The gain is the information the evolution of the bias would have to supply, asserted instead by the prior: where the model keeps the freedom the bias has on the wide original samples, a constant removes even the freedom the wide sample's own evolution needs. Even $\Delta z = 0.2$, one or two bins per sample, gains a factor of 3.6 to 7.1 and fails the test: giving up the tilts and the residuals there gives up real freedom.

What the bottom row tests. The random truths change only the linear bias, so the bottom row tests the linear bias, not the higher-order terms. The four versions that free $b_{1,i}$ per bin sit exactly at 1 by construction: a free per-bin linear bias absorbs any change of the linear bias, so their honesty there says nothing about their higher-order freedom. The version with correlated residuals, whose b_1 keeps the model's prior, stays between 0.98 and 1.00, and the model itself at most 1.018 (Section 5).

7 Fisher forecast and its numerical convergence

The Fisher matrix. Let $\mathbf{m}(\mathbf{p})$ be the model band powers at the parameters $\mathbf{p}$, $\mathbf{p}_0$ the fiducial point, C the band-power covariance, and $\mathsf{D} = \partial\mathbf{m}/\partial\mathbf{p}$ at $\mathbf{p}_0$ (sans-serif, unlike the growth factor D). At a scale cut that keeps the set K of band powers, the Fisher matrix of a fit is

$$F \;=\; \mathsf{D}_K^{\mathsf{T}}\,\mathsf{C}_{KK}^{-1}\,\mathsf{D}_K \;+\; F_{\mathrm{prior}}, \tag{11}$$

with D_K the kept rows of D and C_{KK} the kept block of C. F_{prior} adds $1/\sigma_{\mathrm{pr}}^2$ to the diagonal for a Gaussian prior of width σ_{pr} and nothing for a flat one. For each group and each quantity that

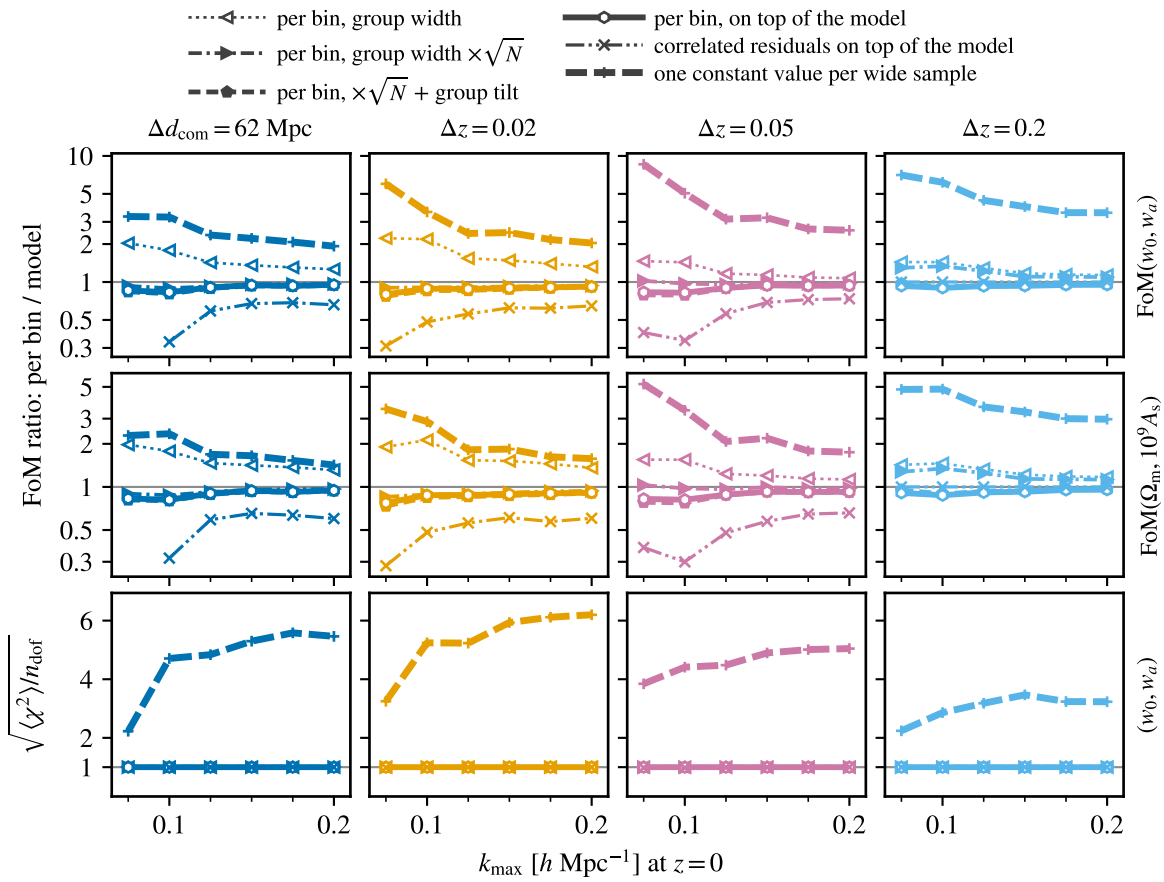


Figure 4. Six versions of the bias priors against the model, one column per layout. Top and middle rows: FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$, each divided by the model's. Bottom row: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of (w_0, w_a) over the random bias histories of Section 5; for the constant version the truth also holds the bias's own evolution inside each sample. The group width is the width the model gives a group parameter, and N the number of bins of the group. The first four versions free $b_{1,i}$ in each bin and give Δ_2, Δ_s, α_0, α_2, SN$_0$ and SN$_2$ one value per bin. Dotted, open left triangles: the group width; dash-dot, filled right triangles: the group width times $\sqrt{N}$; dashed, filled pentagons: the same plus a free tilt per group; solid, open hexagons: the group width on top of the model's offsets and tilts. The line width grows in that order. Thin dash-dot-dot line, crosses: the model, its b_1 prior included, plus correlated per-bin residuals on Δ_2, Δ_s, α_0 and α_2; on $\Delta d_{\rm com} = 62$ Mpc it has no well-defined fit at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$, hence the gap. Thick long-dashed line, plus signs: one constant value of every bias quantity per wide sample. Thin gray line: a ratio of 1, and honest error bars in the bottom row. There the other five versions coincide at 1; the solid line, drawn on top, hides all but the crosses. The cuts start at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$: at $0.05\,h\,{\rm Mpc}^{-1}$ the model's own FoM(w_0, w_a) is below 0.04. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

carries a residual (Table 1) it adds the precision $\mathsf{P} = \mathsf{Q} - \mathsf{QU}(\mathsf{U}^\mathsf{T}\mathsf{QU})^{-1}\mathsf{U}^\mathsf{T}\mathsf{Q}$ on the bins' deviations from the group value (δ_i for b_1), with Q the inverse kernel matrix of Eq. (10) and U the matrix with columns 1 and $\bar{z}_i - \bar{z}_g$: it charges only the residual around the group's best straight line. The marginalized error of parameter j is $\sigma_j = [(F^{-1})_{jj}]^{1/2}$, and the FoMs of Section 2 are $[\det \mathsf{C}_p]^{-1/2}$, with C_p the 2×2 block of F^{-1} of the two parameters. Where a cut leaves a combination of nuisance parameters without information, for instance the flat tilts of a group most of whose bins have lost their bands, F is inverted as a pseudo-inverse: scaled to unit diagonal, diagonalized, and restricted to the eigen-directions above 10^{-12} of the largest eigenvalue. A dropped direction may not touch

(w_0, w_a), Ω_m or A_s by more than 10^{-6}; otherwise the cut gives no fit.

Derivatives. The columns of D come from three kinds of derivatives at the fiducial point.

1. The cosmological parameters, the σ_v and the per-bin $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$, on which the band powers depend nonlinearly, take the five-point central stencil

$$\frac{\partial \mathbf{m}}{\partial \theta} \simeq \frac{\mathbf{m}(\theta_0 - 2\epsilon) - 8\,\mathbf{m}(\theta_0 - \epsilon) + 8\,\mathbf{m}(\theta_0 + \epsilon) - \mathbf{m}(\theta_0 + 2\epsilon)}{12\,\epsilon}, \tag{12}$$

   with θ_0 the fiducial value and $\epsilon = h\,\theta_0$, $h = 0.02$ (an absolute step where $\theta_0 = 0$); its truncation error is of order ϵ^4. Every evaluation is a full call of the likelihood, CAMB included.

2. SN_0, SN_2 and α_2 enter linearly, so their columns are exact: the projected templates per unit amplitude for the stochastic terms (Eq. (9) for SN_2), and a two-point difference for α_2.

3. The per-bin columns are computed in batches. A spectrum C_ℓ^{ij} depends on the biases of bins i and j only, so the bins of a set $\mathcal{S}$ in which no two bins share a measured spectrum can be stepped together. With $\partial_{\mathcal{S}}\mathbf{m}$ the stencil of Eq. (12) with the bias x stepped in every bin of $\mathcal{S}$,

$$\frac{\partial m_r}{\partial x_i} = \begin{cases} (\partial_{\mathcal{S}}\mathbf{m})_r & \text{if band power } r \text{ belongs to a spectrum that contains bin } i \in \mathcal{S}, \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

   A greedy coloring of the graph of measured pairs gives four such batches on every layout, so each per-bin family costs 16 evaluations, whatever the number of bins.

One base, every model. The columns of one layout, for the cosmology, the σ_v and the seven per-bin families b_1, b_2, b_{s^2}, α_0, α_2, SN_0 and SN_2, form a base matrix B, computed once and cached. A model is a fixed linear map from its parameters p_j to the base quantities κ_a,

$$\mathsf{D} = \mathsf{B}\,\mathsf{M}, \qquad \mathsf{M}_{aj} = \frac{\partial \kappa_a}{\partial p_j}\,: \tag{14}$$

a group offset puts ones on its group's bins in its family's block, a tilt the weights $\bar{z}_i - \bar{z}_g$, a bin's linear bias the chain rule through the calibrations of Eq. (8), and the untied versions of Section 6 per-bin identities. With $\mathsf{G}_K = \mathsf{B}_K^{\mathsf{T}} \mathsf{C}_{KK}^{-1} \mathsf{B}_K$ computed once per cut, every model and prior at that cut has $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_K \mathsf{M} + F_{\text{prior}}$: a change of model, prior, cut or truth needs no new evaluation of the data vector.

Shifts and the error-bar test. A truth that departs from the fitted model moves the band powers by $\Delta\mathbf{m}$, and the best fit, to first order, by

$$\delta\mathbf{p} = F^{-1}\,\mathsf{D}_K^{\mathsf{T}}\,\mathsf{C}_{KK}^{-1}\,\Delta\mathbf{m}_K, \tag{15}$$

exactly for the terms linear in their amplitude (α_2, SN_0, SN_2) and to first order for the biases, on which the band powers depend quadratically. Because $\delta\mathbf{p}$ is linear in the truth, the average behind the error-bar test of Section 5 is exact in closed form; for (w_0, w_a),

$$\langle \chi^2 \rangle = \Delta^2_{\text{drift}} + \operatorname{tr}\big[\mathsf{C}_p^{-1}\big(\mathsf{A}\,\Sigma\,\mathsf{A}^{\mathsf{T}} + \mathsf{N}_p\big)\big], \tag{16}$$

with Δ_{drift} the shift of the known drift alone in units of the ellipse, A the response of (w_0, w_a) to the bins' deviations δ_i, Σ the covariance of the residuals drawn from the prior, and N_p the (w_0, w_a) block of $F^{-1} F_0 F^{-1}$, the share of the data noise and the other priors, F_0 being F without the correlated prior. For Ω_m or A_s the same holds with the parameter's own row.

Numerical convergence. Noise in the data vector, from its integration grids and CAMB's tolerances, enters a finite difference divided by the step. A column $\mathbf{d}_j + \mathbf{e}_j$ with noise $\mathbf{e}_j$ raises F_{jj} on average by $\langle \mathbf{e}_j^\mathsf{T} \mathsf{C}^{-1} \mathbf{e}_j \rangle > 0$: noise reads as information and inflates the FoM. Settings that converge a χ^2 to the standard of MCMC runs, $|\Delta\chi^2| < 0.2$, are therefore not enough. On $\Delta z = 0.05$ they inflate $\mathrm{FoM}(w_0, w_a)$ by 58% at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and 11% at $0.2\,h\,\mathrm{Mpc}^{-1}$, and the earlier derivatives inflated it about threefold on $\Delta d_{\mathrm{com}} = 62$ Mpc at $0.075\,h\,\mathrm{Mpc}^{-1}$; the excess comes from noise in the w_0, w_a and Ω_m columns. That spurious information also weakened the degeneracy between α_2 and the cosmology, so the earlier derivatives understated the α_2 shifts of Figure 3. The criterion of convergence is step independence: the FoMs, the errors and the shifts must not move when the stencil step changes or the numerical controls are refined further. A study on $\Delta z = 0.05$ (Figure 5) starts from a configuration with every control raised: the grid factor $g = 128$, where g counts the comoving distances of the non-Limber integrals in units of 511; CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2; the low-rank engine's rank 8 and multipole cap 2000; and CAMB's `AccuracyBoost` 2, `k_per_logint` 30 and maximum wavenumber 80 Mpc^{-1}. From there it changes one control at a time and compares each configuration with the converged value, the five-point stencil at step h on $g = 512$. The final settings are:

- the final grid $g = 256$, eight times the MCMC grid of $\Delta z = 0.05$, with CosmoLike's `accuracyboost` 3 and `integration_accuracy` 1;
- CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and a maximum wavenumber of 60 Mpc^{-1}, with halofit's bisection tolerance tightened from 10^{-3} to 10^{-7} (the halofit tolerance patch);
- five-point stencils for every stencil and per-bin column.

Against the converged value they give $\mathrm{FoM}(w_0, w_a)$ +1.8% at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and −0.5% at $0.2\,h\,\mathrm{Mpc}^{-1}$ ($\mathrm{FoM}(\Omega_m, 10^9 A_s)$ +1.6% and −0.5%), marginalized errors within 1.6% and shifts within 0.007σ. From step $h/2$ to $3h$ both FoMs stay within 1.6% on $\Delta z = 0.05$ and within 0.3% on $\Delta d_{\mathrm{com}} = 62$ Mpc. Two cross-checks agree. DerivKit, which fits a polynomial through at least seven evaluations around the fiducial point, matches the stencils within 2.1%. The study's route, which unties the per-bin parameters in the likelihood, matches the production route, which steps them through CosmoLike's setter: exactly for b_2, b_{s^2} and α_0, within 5×10^{-11} for b_1 and within 5×10^{-14} for the stencil columns.

Cost. A layout's base takes 156 evaluations of the data vector, about 20 to 40 s each at the final settings, so one to two hours per thin layout; every model, prior, cut and truth then costs a fraction of a second. `notes/fisher_machinery` gives the details.

8 Where the details are

- `notes/correlated_bias_prior_v2`: the linear-bias prior and its physics, the projected precision, the random bias histories and the error-bar test, and the comparisons of priors and models.
- `notes/higher_order_bias`: the one-loop model, its calibrations, the size and exact projection of every term, and its validation.
- `notes/linear_bias_kmax`: the reach of linear bias, which motivates the one-loop terms.
- `notes/fisher_machinery`: the Fisher derivatives, their convergence study, and the assembly of every Fisher matrix, shift and $\langle \chi^2 \rangle$.

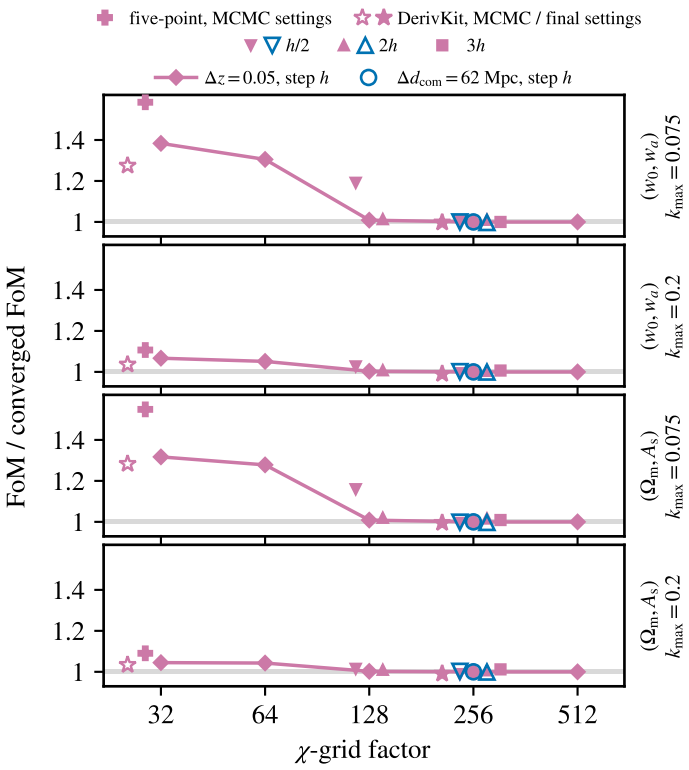


Figure 5. Convergence of the forecast on $\Delta z = 0.05$: the FoMs of the model divided by their converged values (five-point stencils at step h on the grid factor $g = 512$). From the top, FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$ at $k_{\max} = 0.075$ and $0.2\,h\,\mathrm{Mpc}^{-1}$. Diamonds on the line: the scan in g at step h, which holds every other control at the study's starting configuration ($g = 128$ with the other controls raised; Section 7); $g = 256$ is the final setting. Filled down-triangles, up-triangles and squares beside a grid: steps $h/2$, $2h$ and $3h$. Plus: five-point stencils at the MCMC settings. Open and filled stars: DerivKit at the MCMC and at the final settings. Open blue markers: $\Delta d_{\mathrm{com}} = 62$ Mpc at the final settings, steps $h/2$ (down-triangle) and $2h$ (up-triangle) against its own step-h value (ring). Gray band: $1 \pm 1\%$. $k_{\max}$ at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

- `notes/cross_correlations`: which thin-bin pairs the data vector measures, and the Fingers-of-God blur.

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